

Problems for week 11

(Questions on this problem sheet are NOT assessed)

Problem 1. Verify that Gauss' Divergence Theorem is true for the vector field $\vec{F}$ on the region D .

- (a) $\vec{F}(x, y, z) = y^2 z^3 \hat{i} + 2yz \hat{j} + 4z^2 \hat{k}$, D is the solid enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 9$.
- (b) $\vec{F}(x, y, z) = (z, y, x)$, D is the solid ball $x^2 + y^2 + z^2 \leq 16$.

Problem 2. Use Gauss' Divergence Theorem to calculate the surface integral

$$\iint_S \vec{F} \cdot d\vec{S};$$

that is, calculate the flux of $\vec{F}$ through S .

- (a) $\vec{F}(x, y, z) = 3xy^2 \hat{i} + xe^z \hat{j} + z^3 \hat{k}$, and S is the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and the planes $x = -1$ and $x = 2$.
- (b) $\vec{F}(x, y, z) = (2x^3 + y^3) \hat{i} + (y^3 + z^3) \hat{j} + 3y^2 z \hat{k}$, and S is the surface of the solid bounded by the paraboloid $z = 1 - x^2 - y^2$ and the xy -plane.

Problem 3. Prove the following identity, assuming that S satisfies the conditions of Gauss' Divergence Theorem and that the components of the vector field $\vec{F}$ have continuous second-order partial derivatives:

$$\iint_S \text{curl}(\vec{F}) \cdot d\vec{S} = 0.$$

Problem 4. Verify that Stokes' Theorem holds true for the given vector field $\vec{F}$ and surface S :

- (a) $\vec{F}(x, y, z) = -y\hat{i} + x\hat{j} - 2\hat{k}$, and S is the cone $z^2 = x^2 + y^2$, $0 \leq z \leq 4$, oriented downward.
- (b) $\vec{F}(x, y, z) = y\hat{i} + z\hat{j} + x\hat{k}$, and S is the hemisphere $x^2 + y^2 + z^2 = 1$, $y \geq 0$, oriented in the direction of the positive y -axis.

Problem 5.

- (a) Calculate the curl of the vector field $\vec{F} = -2xz\hat{i} + y^2\hat{k}$.
- (b) Using Stokes' Theorem, show that

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

for any simple closed curve C drawn on the unit sphere $x^2 + y^2 + z^2 = 1$.